



Artificial Intelligence and Machine Learning

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Department of Computer Applications

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AI Landscape

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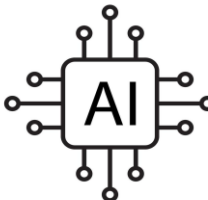
Dictionary meaning

Dictionary
Definitions from [Oxford Languages](#) · [Learn more](#)

 **intelligence**
/ɪnˈtɛlɪdʒ(ə)ns/

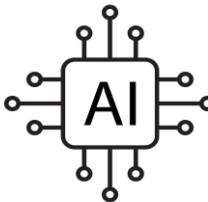
noun

1. the ability to acquire and apply knowledge and skills.
"an eminent man of great intelligence"

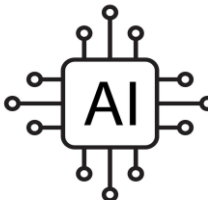
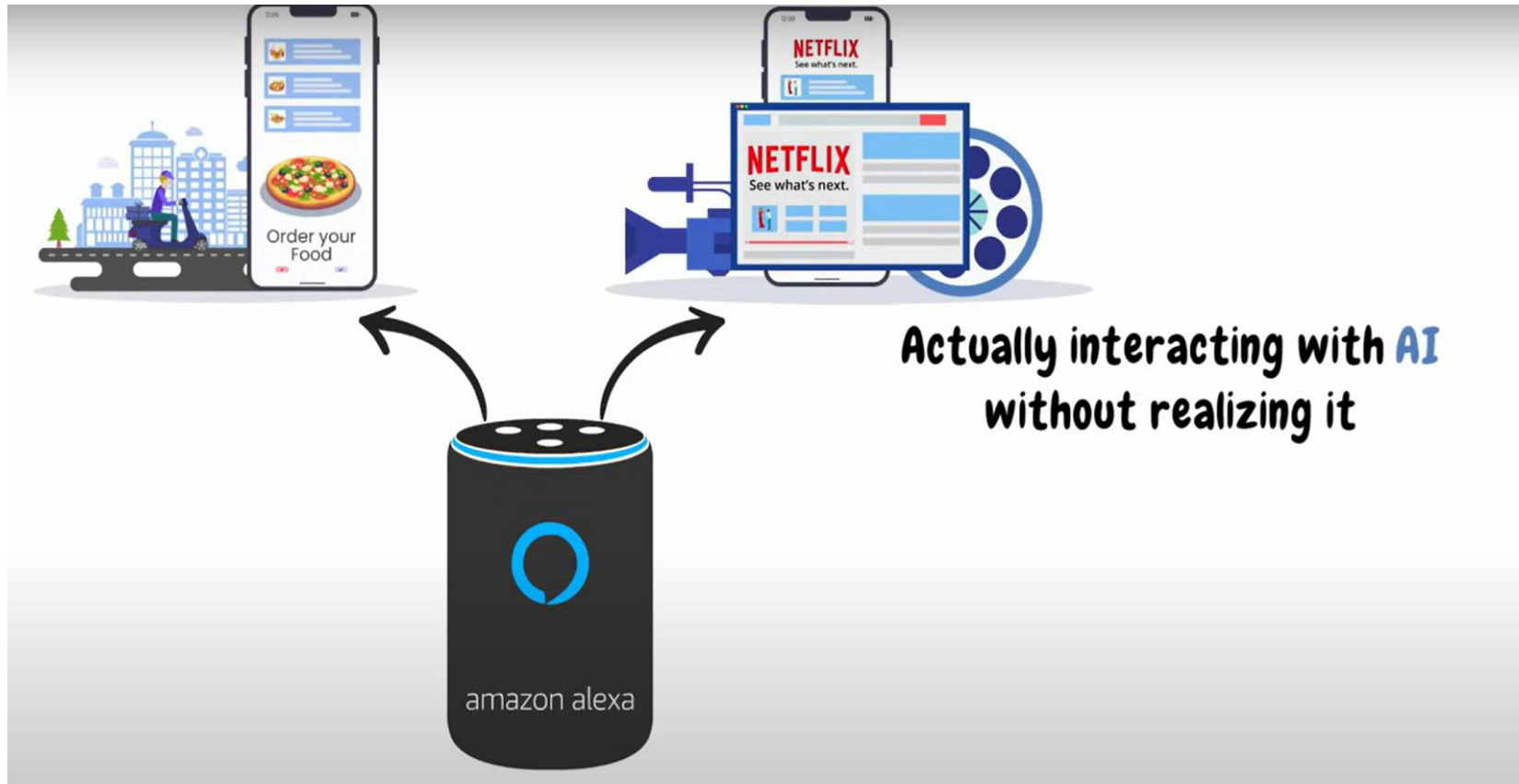


Artificial

Intelligence

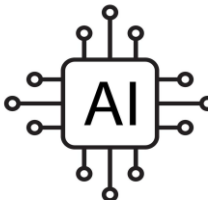
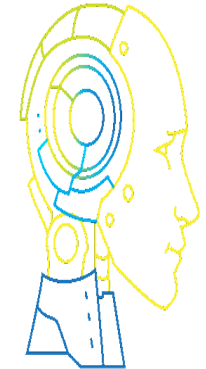


Computer Scientists inspired by the human brain processes imparted the intelligence to machines – **Termed as AI**



Definition

- The area of computer science that studies how machines can closely imitate human intelligence
- Make computers to “think” like a human
- Reflect some systems which brings the mechanics of the brain within computing
- It is a branch of computer science by which we can create intelligent machines which can **behave like a human, think like humans, and able to make decisions.**



History of AI

1950 – Alan Turing speculated about the possibility of creating machines that think

1951 – Game AI – checkers program

1956 – John McCarthy coined the word AI in a conference

1959 – First AI Lab at MIT

1960 – Robot in assembly line by General Motors

1961 – An AI chatbot ELIZA was introduced

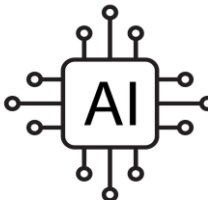
1997 – IBM's Deep Blue – beats Gary Kasparov in chess

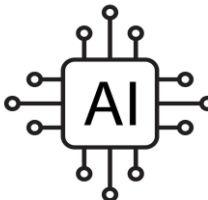
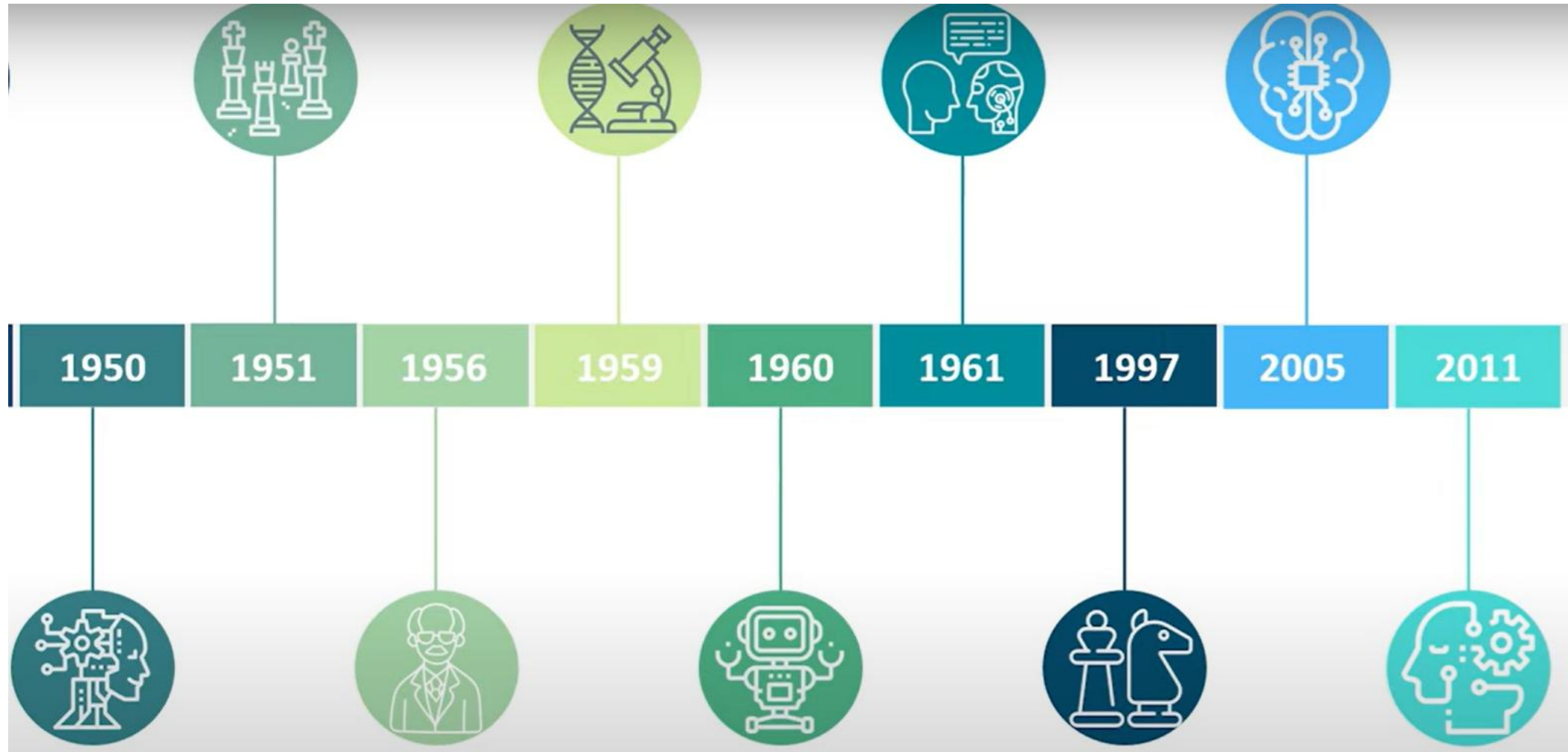
2005 – Autonomous car named Stanley wins DARPA Grand Challenge

2011 – IBM's Watson - QA System



Alan Turing



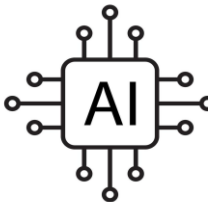




John McCarthy - coined the word Artificial Intelligence in 1956

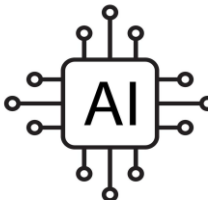
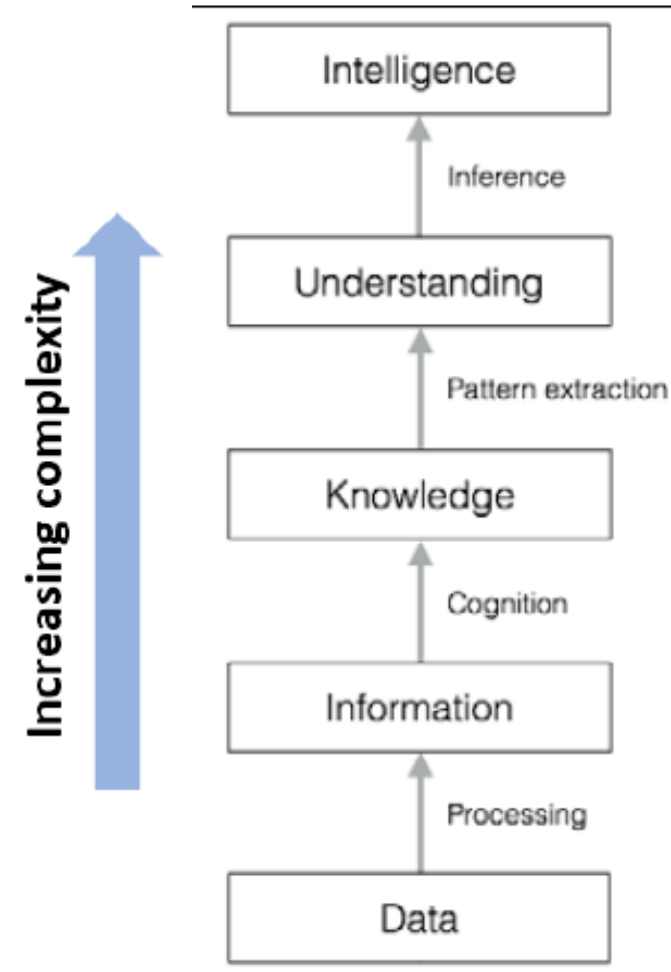
“AI is the Science and Engineering of making Intelligent Machines”

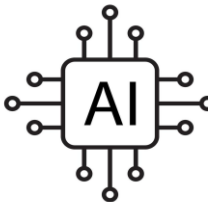
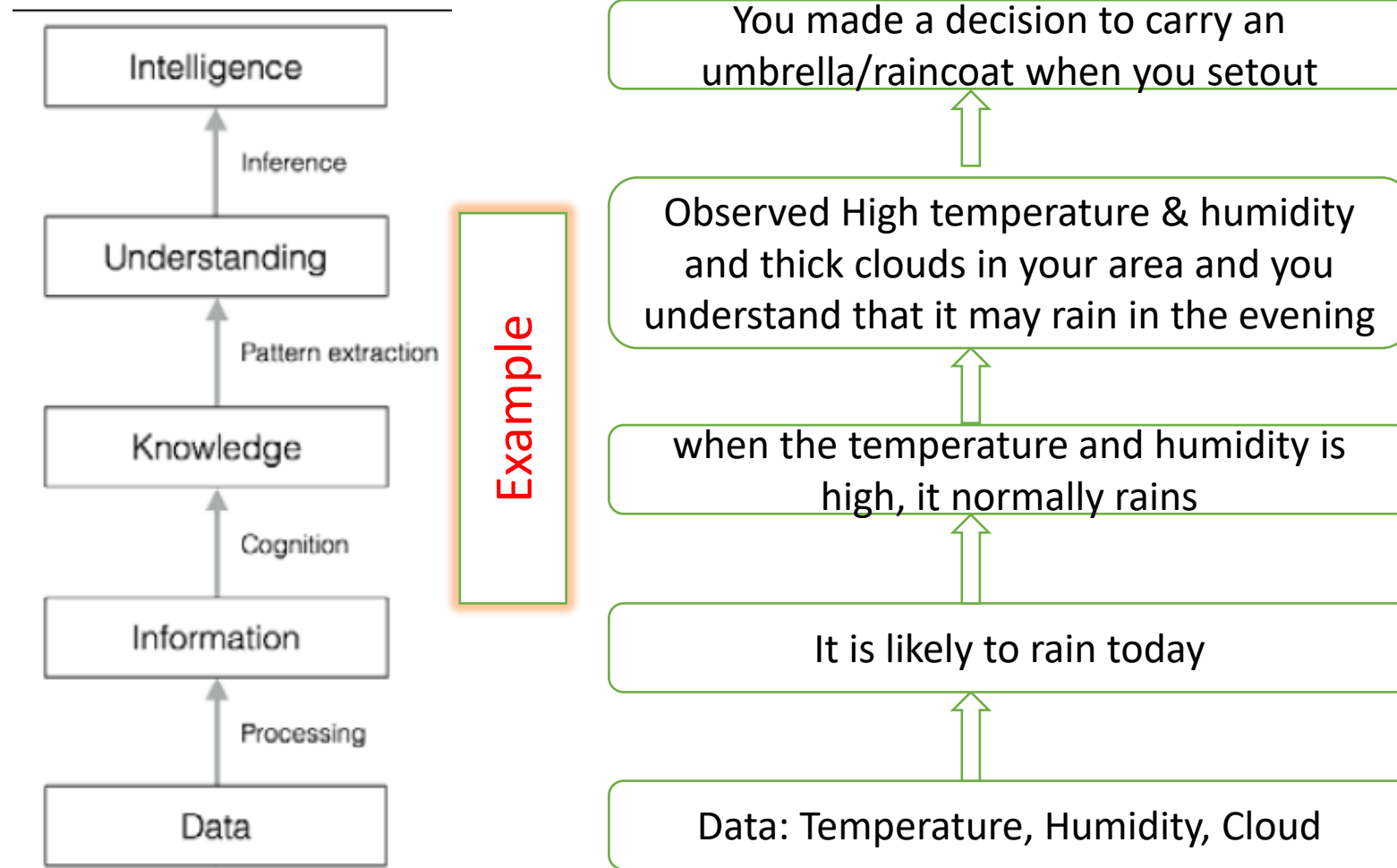
- John McCarthy



Raw Data into actionable intelligence

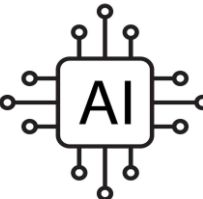
- We deal with huge amounts of data. The human brain can't keep track of so much data
- Data originates from multiple sources simultaneously. The data is unorganized and chaotic
- Process the data to make meaningful information
- Interpret the information that helps decision-making
- Understand the patterns, meanings and implications
- Ability to predict and make decisions





Need to Study AI

- AI is needed to automate tasks, analyze complex data, improve decision-making, and solve problems that are difficult or impossible for humans alone
- To make advancements in various fields like healthcare, manufacturing, and research
- AI is needed for
 - Automation and Efficiency
 - Enhanced Decision Making
 - Solving Complex Problems
 - Social Benefits





THANK YOU

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